# An audit of the 2026 Navier–Stokes finite-time blowup claim: the formal statement read against Clay's wording and Mathlib's definitions, the Lean proof built and its axioms asked of the kernel, and what remains open

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October 2026 — draft v0.2 for review (v0.1 read the announced commit only; v0.2 adds the upstream commit of 2026-09-10, read here 2026-10-05); every number interpolated from the record; machine-derived, not peer-reviewed; author list to be agreed

**Abstract**

On 8 September 2026 OpenAI published a 166-page proof, with a Lean 4 formalisation, that for every positive viscosity a smooth compactly supported force drives the three-dimensional incompressible Navier–Stokes flow from rest to unbounded velocity in finite time, and stated that this establishes alternatives (C) and (D) of the Clay problem statement [4, 2, 1]. This note records an audit made the following day from a corpus pinned by digest: the formal statement read hypothesis by hypothesis against Fefferman's wording and Mathlib's definitions; the 618,762-line repository built on a laptop and the kernel asked for the axioms of each main declaration; Comparator run with a second kernel; the paper mapped onto the certificate clause by clause; the writeup dissected by six independent readings and set against every classical obstruction; every printed identity a computer can decide re-run with red controls. What holds: the statement is Clay's with nothing added or dropped, the build passes on the three standard axioms, both kernels accept, and no error or counterexample was found. What the audit found at the announced commit, `8937a8f4`: the paper's finite-energy clause had no formal counterpart, the lemma that gives it proved and used by nothing. The one upstream commit since, `f9e8bc5b` of 2026-09-10, the day after the audit, fills that clause from that lemma and proves the paper's Theorem 1.1 as stated; this note did not say so until 2026-10-05. It was rebuilt here over the pinned build and the kernel reports the same three axioms for all 7 declarations asked. Comparator was not re-run at that commit, and the statement of Theorem 1.1 it proves is OpenAI's own, not a third party's: its fidelity rests on this note's clause-by-clause reading. Not decided: the smallness parameter is bounded numerically nowhere; 3 checks of the companion Euler paper are listed and unrun. The register row reads CERTIFIED, recorded 2026-10-05, with the PARTIAL of 2026-09-09 at `8937a8f4` kept as its history. Nothing is claimed about the mathematics beyond what the kernel and the readings establish, nor about priority.

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

# 1 Introduction

The claim is Theorem 1.1 of [2]: for every $\nu > 0$ there are a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$, a compact $K \subset \mathbb{R}^3$ and smooth $(u, p)$ on $\mathbb{R}^3 \times [0, 1)$, supported in $K$, solving the Navier–Stokes equations from rest, with $\sup_{0 \le t < 1} \|u(t)\|_{L^2} < \infty$ and $\limsup_{t \uparrow 1} \|u(t)\|_{L^\infty} = \infty$; consequently no smooth solution on $\mathbb{R}^3 \times [0, \infty)$ with the same force and datum has uniformly bounded kinetic energy. The paper says this is alternative (C) of the Clay problem statement [1] and, through its Corollary 10.6, alternative (D). A Lean 4 repository [5] formalises both; the announcement [4] says the proof was produced by an internal model and formalised by another.

A formal certificate changes what an audit of such a claim can be: three questions become decidable by machine and one by reading. Is the formal statement the theorem the prose names; does the proof term check, and on which axioms; does the paper say what the certificate says; and where the paper asserts rather than displays, does the certificate discharge the assertion? The position throughout is the one the records state: *the Lean certificate is the claim; the paper is its human-readable evidence.* A gap in the paper is a gap in the writeup unless the Lean shows the same, and a printed identity that failed would refute the formula, not the theorem [19]. The contribution is not a theorem but the grammar of the audit and its public record; every number below is interpolated from a record by a tool that refuses when a record no longer supports a sentence.

The audit read `8937a8f4`, the commit the announcement shipped, on the day after it appeared. Upstream moved once afterwards, to `f9e8bc5b` on 2026-09-10, and the record said nothing about that commit until 2026-10-05. The audit's findings are kept as findings about the commit they read, dated; the later commit is read at the same level in a section of its own, and the register's verdict follows it with the earlier verdict kept as history. Sections 2–7 give the claim and the corpus, the audit, the upstream commit, what the record decides, what is not claimed, and the dates as the parties state them, weighed by nobody here.

# 2 What was claimed, and what was pinned

**Clay's alternatives.** Fefferman's statement [1] asks for a proof of one of four assertions; in its wording, (C) *Breakdown of Navier–Stokes solutions on* $\mathbb{R}^3$: take $\nu > 0$ and $n = 3$; then there exist a smooth, divergence-free vector field $u^\circ(x)$ on $\mathbb{R}^3$ and a smooth $f(x,t)$ on $\mathbb{R}^3 \times [0,\infty)$, satisfying (4), (5), for which there exist no solutions $(p, u)$ of (1), (2), (3), (6), (7) on $\mathbb{R}^3 \times [0,\infty)$; and (D) the same on $\mathbb{R}^3/\mathbb{Z}^3$ with (8), (9) in place of (4), (5) and (10), (11) in place of (6), (7), the errata adding that the pressure is periodic. The numbered conditions are those of Table 1.

**The mechanism.** As the records describe it [19]: a self-similar axisymmetric core with radial scale $\tau^{1/2}$, axial scale $\tau^{1/2-h}$ and azimuthal and axial speeds $\tau^{-1/2-h}$, $\tau = 1 - t$, $0 < h < 1/100$, whose momentum residual is unbounded in an annulus; two families of shear-amplified oscillatory pulses on an auxiliary torus that supply the missing stress through their averaged quadratic flux; corrections at every order in $q^{2h}$; an exact radial-heat exterior; cutoffs. The force is *defined* as the residual of the constructed flow and shown to extend smoothly through $t = 1$; uniqueness among smooth finite-energy solutions with a merely smooth pressure transfers the growth to any competitor, in the forced-blowup program of Córdoba and Martínez-Zoroa [9].

**The corpus.** Everything read was pinned on 2026-09-09 with its digest: 24 files: the two papers, the Clay statement, the Lean bridge files, the Formal Conjectures statement, the announcement and the concurrent papers and statements [2, 3, 4, 5, 6, 10]. When pinned, the repository [5] had one commit, `8937a8f4`, at 2026-09-08 10:57 UTC (a second followed on 2026-09-10; Section 4); it pins Lean v4.34.0-rc2 [7] and Mathlib at `85e3a25e` [7]; at that commit it holds 2,486 files and 618,762 lines, of which 404,869 lines in 644 files are the Navier–Stokes library, 213,419 in 1,840 the Euler library and 474 in 2 the challenge statements. Its `formalization.yaml` declares the review *self-assessed* and the automation *GPT-6 Astra*, with no `sorry` [19].

# 3 The audit

## 3.1 The statement, hypothesis by hypothesis

The theorem the certificate proves was not written by the prover. The Comparator challenge module is the Formal Conjectures statement of the Clay alternatives [6], added by DeepMind on 2026-05-15 (pull request #1457), last touched on 2026-07-27 at revision `8bf45ed7`, the revision OpenAI pinned and the revision on `main` on the day of the audit. OpenAI's copy differs from upstream in 4 lines, every one an `open` or a notation, and the solution side re-declares the same definitions under the same names, byte-identical once comments are stripped [19]. Comparator compares every constant reachable from the theorem's type for exact equality, so the only question is whether that statement is Clay's.

Both theorems have the shape $\exists\, u_0\, f,\ H(u_0) \wedge H(f) \wedge \neg\exists\, v\, p,\ S(v,p)$: the prover must *establish* $H$ for the constructed data and *exclude* every $(v,p)$ in $S$. A junk value or a weak definition inside $S$ only enlarges the class and strengthens the theorem; a gap could hide only in an $S$ stronger than Clay's (6)(7) or (10)(11), excluding competitors Clay admits, or in an $H$ weaker than Clay's (4)(5) or (8)(9), admitting data Clay rejects. Every row of Table 1 is read in that direction, with the Mathlib definition located at the pinned revision [19].

**Proposition 1** (statement fidelity; the record's verdict)**.** *The formal statements of (C) and (D),* `navier_stokes_breakdown_R3` *and* `navier_stokes_breakdown_periodic`*, are Clay's with no hypothesis added to the competitor class and none dropped from the data; in particular no decay, growth or integrability condition is imposed on the pressure or on $\nabla v$ beyond smoothness, so the formal uniqueness argument must handle every competitor Clay admits. The dimension is* `Fin 3` *and $\nu$ is any positive real.*

A negation is cheap if the class it negates is empty for a trivial reason, so a witness was written for this audit and compiled against OpenAI's challenge module: the fluid at rest satisfies every field of both competitor structures and the zero data all four data conditions (`cert-machine-NonVacuity.lean`, 7 declarations, 0 errors, the three standard axioms for `classes_inhabited`) [19]. It shows the definitions are satisfiable, not that they admit every classical solution, which is Table 1's job.

## 3.2 The build, the axioms and the second kernel

The repository was built on 2026-09-09 on an ordinary laptop (Mac14,2, 8 cores, 16 GB, darwin 25.4.0, arm64) with the pinned toolchain: all 2,486 modules in 140 minutes after the Mathlib cache (11,251 Lake jobs in all; two legs, 681 modules at 8 threads then 1,805 at 4, after eight parallel Lean processes drove the machine into swap), exit 0, no errors [19]. A scratch file importing

| Clay | Lean (Formal Conjectures) and the Mathlib definition at `85e3a25e` | direction |
|---|---|---|
| (1) $\partial_t u + (u \cdot \nabla)u = \nu\Delta u - \nabla p + f$ on $\mathbb{R}^3 \times [0,\infty)$ | `derivWithin (v x ·) (Ici 0) t + fderiv ℝ (v · t) x (v x t) = nu • Δ (v · t) x - gradient (p · t) x + f x t`, $\forall x$, $\forall t \geq 0$: the one-sided time derivative at $t = 0$ within $[0,\infty)$; `fderiv` applied to $v$ is $(v \cdot \nabla)v$; $\Delta$ the ordinary Laplacian (trace of the second derivative); `gradient` the Riesz representative of `fderiv` | same |
| (2) $\operatorname{div} u = 0$ | $\nabla\cdot$ `(v · t) x = 0`, the trace of the Jacobian | same |
| (3) $u(x,0) = u^\circ$ | `v x 0 = u₀ x` | same |
| (6) $p, u \in C^\infty(\mathbb{R}^3 \times [0,\infty))$ | `ContDiffOn ℝ ∞ (↿v) (univ ×ˢ Ici 0)`, likewise $p$: $C^\infty$ up to $t = 0$ in the Taylor-series sense, not analytic | same |
| (7) $\int \lvert u\rvert^2 dx < C$ for all $t$ | `integrable : ∀ t ≥ 0, MemLp (‖v · t‖) 2` with `globally_bounded_energy : ∃ E, ∀ t ≥ 0, ∫ ‖v x t‖² < E`; the Bochner integral of a non-integrable function is 0, which the first field forbids | same |
| (10) $u$ periodic; errata: $p$ periodic; (11) smooth | `IsOnePeriodic (v · t)`, `IsOnePeriodic (p · t)` for $t \geq 0$, along `EuclideanSpace.single i 1`; smoothness as (6) | same as Clay plus the errata |
| (4) $u^\circ$ smooth, divergence-free, $\lvert\partial^\alpha u^\circ\rvert \leq C(1+\lvert x\rvert)^{-K}$ $\forall \alpha, K$ | `InitialVelocityConditionDecay`: divergence-free, `ContDiff ℝ ∞ u₀`, $\forall m\, K\ \exists C$, $\|$`iteratedFDeriv ℝ m u₀ x`$\| \leq$ `C/(1+‖x‖)^K`; the operator norm of the $m$-th derivative is comparable to the largest order-$m$ partial | equivalent; the paper takes $u^\circ = 0$ |
| (5) $f$ smooth on $\mathbb{R}^3 \times [0,\infty)$, $\lvert\partial_x^\alpha \partial_t^m f\rvert \leq C(1+\lvert x\rvert+t)^{-K}$ | `ForceConditionDecay`: `ContDiffOn ℝ ∞ (↿f) (univ ×ˢ Ici 0)` and the same bound on `iteratedFDerivWithin` of total space-time order $m$ in place of Clay's split $(\alpha, m)$ | equivalent; a $C_c^\infty$ force satisfies it trivially |
| (8) $u^\circ$, $f$ periodic in $x$; (9) $\lvert\partial^\alpha \partial^m f\rvert \leq C(1+t)^{-K}$ | `InitialVelocityConditionPeriodic`; `ForceConditionPeriodic` with the time-decay bound on `iteratedFDerivWithin` | same |

Table 1: Clay's hypotheses against the formal statement, one row per hypothesis (9 rows; 7 read *same*, 2 *equivalent*). The first six rows are the competitor class $S$, the last three the data hypotheses $H$. Transcribed from the record [19].

the two solution modules then asked `#print axioms` of the 4 declarations `formalization.yaml` names, and the kernel reported exactly `propext`, `Classical.choice`, `Quot.sound` for each. In code lines, comments blanked, the proof libraries contain no `sorry` beyond the 4 intentional placeholders of the challenge files, no axiom declaration, no `native_decide`, `unsafe`, `opaque` or `partial def`, no custom syntax and no heartbeat override (5 docstrings record earlier overrides removed).

Then the judge. Comparator [8], the Lean FRO's tool for exactly this situation, exports both sides with `lean4export`, compares every constant reachable from each theorem's type for exact equality, walks the axioms, and replays the proof through Lean's kernel and through nanoda, a kernel written independently of Lean's in Rust and built from source for this run. On the 2 configurations, Navier–Stokes and Euler, all 4 kernel replays accept and 2 of 2 return *"Your solution is okay!"* with exit 0, in 13 and 17 minutes. One component did not run and the record says so: Comparator's Landlock process isolation is Linux-only and a pass-through stand-in took its place; the statement comparison, the axiom walk and both kernel replays were performed, the isolation was not, and on a machine where both sides come from one pinned commit it guards against nothing this audit needed. What the build establishes is bounded by its trusted base, the two kernels, Mathlib's definitions at the pinned revision, the elaborator and

the hardware: the proof terms exist and check against the third party's statement, and nothing about the paper.

### 3.3 The paper against the certificate

Each clause of the stated theorem was located in the formal library at the pinned commit or named as absent (Table 2; 7 rows in the record, 2 of them not formal at `8937a8f4` and 0 at `f9e8bc5b`, Section 4) [19].

One difference bears on the announcement, which says the fluid's "energy remains finite through the entire dynamics, from rest to the formation of the singularity" [4]. Theorem 1.1 states it and Lemma 10.4 proves it in a few lines, re-derived in the reading. At the announced commit the repository contained the general lemma (`R3/CompactEnergy.lean:343`), nothing invoked it, and `breakdownStatement`, the file's own "full assertion of Theorem 1.1", was never proved. Clay's (C) does not need the clause, since a global smooth competitor is excluded by uniqueness on every $[0, T]$, $T < 1$, where the candidate's energy is finite by compact support alone; so at that commit the formal result stood as Clay's and the physically striking clause rested on the paper. The Euler theorem in the same repository carries its energy bound (Section 3.8); the record read the asymmetry as an oversight rather than an obstacle, as a reading. The reading held: the upstream commit of the next day fills exactly that field with exactly that lemma and proves `breakdownStatement` (Section 4).

### 3.4 The writeup, read

The paper was taken apart the day after it appeared by six independent readings, five section memos (§3 with §10, §4 with Appendix B, §5 with Appendix A, §6 with §7, §8 and §9 with Appendix C) and one map of the Lean library against all of them, each from the pinned PDF, written in the machine-assisted setting the author's disclosure describes and consolidated into one record [19]. Each reading transcribed the statements, traced the order in which every parameter is fixed and re-derived what it could; no circularity and no error were found. What was verified by hand holds as printed in all 6 groups of the record: the closing argument of §10 term by term; the leading equations of §4 re-derived in the similarity coordinates; the heat exterior of Appendix A with Lemma A.8's two zero-integral identities; the identities of §§6–7; the moment system of §8 and the loop of Lemma C.1 on 371 random relaxed-cone data, where choosing $\delta_L$ from the base point alone breaks the cone in about 7% of trials, so the ordering "$\delta_L$ after $\mu_{\max}$, uniformly" is load-bearing; and the coefficient equations of §5, by computer (Section 3.7).

What the paper asserts rather than displays was listed and ranked by how much rests on it, 13 points in all [19]. Point 0 is the subject of Section 3.6; the other 12, in the record's order, are the endpoint regularity of the local field up to $\tau = 0$ in Proposition 9.9 (one paragraph, no estimate displayed, on which the force's smoothness at $t = 1$ rests); the uniformity in $\eta$ of the flatness (3.4) that (10.9) needs; the admissibility of next-cycle sources in Proposition 9.3(i), the one unstated induction hypothesis of §9; the smallness parameters, every one "sufficiently small" and none numeric, the printed $h < 1/100$ not being the operative bound (Lemma 4.8 needs $h < \min\{\lambda, e^{-T_d}\}$ with $T_d = e^{M_d} + 10$, the derivative scale of §6 exceeds one only from dyadic level about $3 \times 10^8$, and flatness to order $N$ needs more than $1000(N + K_m)$ cycles and about $10^7 N$ integrations by parts); the Lipschitz bounds of Proposition B.2; the inequality $2 + h < a$ of Lemma 4.9 on the terminal collar; the continuation of Proposition B.5; the analyticity region of Lemma 5.1 against Theorem 4.6(i); the coefficient equations (5.3)–(5.6), since discharged

by computer algebra; the two correction routes of §7 and §8.5; the hypothesis $|T_0| \geq c\zeta$ of Proposition C.3, never cited in §9; and the overloading of $q_*$. None is an error. 5 bookkeeping slips were found, each harmless with a positive margin (a dropped $-\kappa_s$ in the zero harmonic of two-wave products has margins 0.08 and 0.4 against $\kappa_s = 10^{-5}$). One is a warning to anyone auditing from extracted text: the cutoff family of Lemma 10.5 on p. 122 extracts as the dilations $\varphi_{4R}, \varphi_{8R}$, under which three inequalities are false, while the glyphs are the powers $\varphi_R^4, \varphi_R^8$, under which every step is exact; no inequality is a finding until its glyphs have been seen.

### 3.5 The asserted steps, traced into the kernel

Each asserted step was traced into the Lean library at the pinned commit and asked a harsher question than "is it discharged": whether the formal statement there is *weaker* than the paper's claim [19]. 6 are discharged by a proved theorem at full strength, 2 of them stronger than the paper: the endpoint regularity is a genuine $C^\infty$ extension across $t = 1$ (`OneSidedExtension`, `JointResidualLimits.lean:72`) and the flatness a joint space-time limit, $\eta$-uniform by construction (`VanishingJointJets`, `JointResidualLimits.lean:84`), exactly the uniformity the reading found missing. 5 are discharged with something altered, among them the smallness parameters: $h$ is not a number in Lean but `Classical.choice` of a nonempty profile datum, constrained by $2h < \lambda < 1/10$ and the hypothesis $h \leq 1/1000$ (`NaturalAxisData.lean:41`); $N$, $\Lambda$, $C$, $q_*$ and the cycle schedule are existentials; the only numeric constants are $\kappa = 10^{-5}$, $\Lambda_{\text{chart}} = 4 - \sqrt{2}$, $X_{\text{big}} = 100$ and $X_i = 110$; the Lipschitz machinery of Proposition B.2 and the analyticity region are proved as machinery, their instantiation for the actual profile not traced to a line. 1 point, Lemma 4.9's collar inequality, has no counterpart, the formal tail being a different plateau construction; 3 of the slips are prose only, and two of the paper's arithmetic slips are absent from the formal object.

Two more answers were read from the source. The construction's non-axisymmetry, which Section 3.6 needs, is a theorem, `angularMode_ne_zero` (`ActualInitialization.lean:52`), not an assumption. And the growth witness, `FinalSlowBase.axis_tendsto`, is the *axial* component on the axis along $t \mapsto (t, 0)$, not the paper's off-axis azimuthal path (10.20)–(10.21); both give $\limsup \|u\|_\infty = \infty$, and no Lean theorem bounds the swirl $r\,u_\theta$ from below.

### 3.6 The adversarial pass

The question an audit of a singularity must ask is whether a theorem says this cannot happen. 9 classical obstructions were set against the construction's own exponents [19]. Of these, 6 are necessary conditions and are met: Serrin's $\int \|u\|_\infty^2 dt = \infty$ [13], with $\|u\|_\infty^2 \asymp \tau^{-1-2h}$; Escauriaza–Seregin–Šverák's $\|u(t)\|_{L^3} \to \infty$ [16], met slowly, as $\tau^{-4h/3}$; the Beale–Kato–Majda integral [15]; Leray's lower bounds [12]; finite energy and finite dissipation, the dissipation rate $\tau^{-1/2-3h}$ being integrable exactly when $h < 1/6$; and the Caffarelli–Kohn–Nirenberg measure bound [14], one point at one time. The other 3 are exclusions and are evaded: the type-I exclusions for axisymmetric flows [18, 17] twice, the flow being type II by the factor $\tau^{-h}$ and not axisymmetric, and the classical no-swirl regularity, the force injecting swirl.

The third is load-bearing and unstated. For an axisymmetric flow the swirl $\Gamma = r\,u_\theta$ satisfies

$$\partial_t \Gamma + u_r \partial_r \Gamma + u_z \partial_z \Gamma = \Gamma_{rr} - \Gamma_r / r + \Gamma_{zz} + r f_\theta,$$

a drift–diffusion equation with no zeroth-order term, derived in the battery from the $\theta$-momentum equation and checked symbolically; so from rest with a bounded compactly supported force,

$\sup|\Gamma(t)| \le \int_0^t \sup|r f_\theta|$ and $|u_\theta| \le C/r$ for every $t < 1$. The paper's leading field has $r\,u_\theta^{(0)} = q^{-h}H$, in its own words on p. 27, unbounded as $q \downarrow 0$. The axisymmetric background alone is therefore impossible as a forced flow from rest; the theorem survives because the physical field is not axisymmetric, the pulses carrying nonzero integer angular frequencies (p. 12) whose Reynolds flux breaks the maximum principle for the angular mean. The paper does not say this anywhere; it is the sentence a referee would ask for, and it is why $h > 0$ is forced rather than chosen. The formal object evades the obstruction by `angularMode_ne_zero` (Section 3.5). This is a reading, not a refutation: the record is explicit that no obstruction refutes the construction.

### 3.7 The printed identities, re-run

The proof carries no numerics and fixes no constant to a number, so there is nothing for an interval certifier to certify. What the paper prints are identities, and each was re-run from the formula as printed by 7 probe scripts carrying 26 red controls, a deliberately wrong variant of each test that must be rejected, 0 of which failed to fire [19]. Symbolically: the similarity-coordinate derivatives of Lemma 4.1, the commutator kernel norm $16\pi/R$ of Lemma 10.5, the derivative count (10.12), the rescalings (10.22)–(10.23) and of Corollary 10.6, the exterior $q$-cancellation, the cutoff bound (10.3), and the exact exterior swirl $K = (r^2/2)^{-A}H(4\tau/r^2)$ against its equation (A.37). At 25 digits: the integral $H$ of (A.32) against (A.37), worst relative residual $4.7 \times 10^{-19}$, and against its Taylor coefficients (A.35) to order 4. By exact algebra: the expansion (5.1) substituted into the axisymmetric operator and compared term by term with the printed coefficient equations (5.2)–(5.6), the sparsity and block nilpotency of $A_1$ in (5.7) and the divisibility of $\Omega_k$ by $X$, 69 identities. In exact rationals: the admissible stress cone of §4.3, where Lemma 4.5's square-root-free form (4.22) agrees with the relaxed condition on a sweep of 25,200 points (7,234 admissible, 17,966 not), a planted sign flip disagrees at 10,456 of them, and Proposition 7.5's printed amplitudes are real exactly on the reference cone at 1,640 points (224 inside); and the swirl equation above. In floating point, deciding nothing, as the record says: the axis initial-value problem of Appendix B at representative data the paper never fixes, whose solved profile tends to the closed form of Proposition B.2 as $1/\Lambda$, with (B.17) holding from $\Lambda = 10^6$ on. The record also keeps this battery's own first bug, a wrong derivative factor ($(-h)_m$ for $(-1)^m(h)_m$) that made a first run report $H$ failing its equation; it is now a red control, so that a scratch log reading FAIL against a Millennium claim is not mistaken for a finding.

### 3.8 The companion theorem

The same repository formalises Theorem 1.1 of [3]: there is $u_0 \in C^\infty_{c,\sigma}(\mathbb{R}^3)$ with $0 < T^*(u_0) < \infty$ whose smooth unforced Euler solution has $\limsup_{t\uparrow T^*}\|\nabla u(t)\|_{L^\infty} = \infty$ and $\int_0^{T^*}\|\operatorname{curl} u(t)\|_{L^\infty}dt = \infty$, a breakdown with no force, read in full [19]. Of its two formal shapes, `exists_compact_smooth_euler_singularity` asserts a nonzero smooth compactly supported datum, $0 < T^* \le 1$, a uniform energy bound on the whole lifespan, maximality as an equivalence, infinite $\limsup$ of the $C^1$ norm and infinite Beale–Kato–Majda integral at $T^*$, and no global smooth finite-energy solution; `euler_breakdown_R3` is the strictly weaker non-existence form, equivalent to blowup for these data only because the formal class imposes square integrability and bounded energy.

The step where such arguments die is the transfer from the approximants $U_j$ to the limiting datum, Euler's stability constant $\exp(C\int\|\nabla U_j\|_\infty)$ being the diverging quantity; the reading finds §6.2 complete as written, because it compares each $U_j$ with the hypothetical smooth

solution of the limiting datum under the contradiction hypothesis $T^*(u_0) > T_\infty$, so the Gronwall constant is $C \sup \|u\|_{H^4}$, fixed before $j$ is chosen. The reading's sharpest candidate error, (3.16) on p. 13 apparently printing $(k \cdot C_*)^{n+1}$ with $C_* = 80$, false at $n = 0$, was WITHDRAWN: the page rendered at 150 dpi shows $(k^{C_*})^{n+1}(n!)^2$; the extraction had flattened the superscript, the second such artifact of this audit. The record lists 3 checks the reading transcribed and did not run: the frozen-frame WKB model as an exact ODE system, every scale inequality of §5.5 and (6.1)–(6.2) in exact logarithmic arithmetic, and the symbolic identities of §3 including the trace check that fixes the $-2$ in (3.14). The Euler result is audited separately and not folded into the Navier–Stokes verdict.

## 4 The upstream commit of 2026-09-10

**What it changes.** The repository's own metadata dates `f9e8bc5b` to 2026-09-10 11:51 UTC, the day after the audit, authored by Boris Alexeev with the message ".". This note learned of it on 2026-10-05 and read it then at the level the audit worked; the computed facts are a record of their own [19]. Between the two commits 188 files changed, 173 added and 15 modified, 25,143 lines in and 81 out, nothing under `Euler/`; the challenge files, the solution-side definitions, `R3/ProblemStatement.lean`, `R3/CompactEnergy.lean` and the toolchain pins are byte-identical to `8937a8f4`. The commit closes both rows of Table 2 that were not formal. The candidate's `energy_bounded` field, left as a goal at `R3/ActualCandidate.lean:114`, is discharged at `R3/ActualCandidate.lean:119` by `CompactEnergy.uniform_finite_energy` (`R3/CompactEnergy.lean:343`, unchanged), the lemma Section 3.3 found used by nothing; the force is now one whose closed support is compact and lies in $t > 0$, which fills `force_support` at `R3/ActualCandidate.lean:110`; and `theorem_1_1 : ProblemStatement.breakdownStatement` is proved at `R3/Theorem.lean:46`. Clay's (C) is now derived from it at `ComparatorR3Theorem.lean:43` through `comparator_of_breakdown` (`R3/ComparatorBridge.lean:77`), and (D) from the periodic corollary at `ComparatorTheorem.lean:52`. In code lines at `f9e8bc5b` there is still no `sorry`, no axiom declaration, no `native_decide` and no `unsafe` across 2,659 files, and `formalization.yaml` still declares the same four main results.

**Built here.** On 2026-10-06, on the machine of the 2026-09-09 build and over its compiled modules, every file the diff touched and everything importing one, 486 modules, was recompiled in 25 minutes, exit 0, no errors; the other 2,173 modules are those built for `8937a8f4`. Lake was not used, and why is a fact about the machine, not the proof: on its current operating system the toolchain's compiled `lake`, `lean4export` and `comparator` abort at start, while `lean` runs clean, so a driver runs the pinned `lean` with the compile line Lake records. A scratch file then asked the kernel, as for the pin, for the axioms of the four declared main results and of `theorem_1_1`, `theorem_1_1_with_dissipation` and `periodic_corollary`: exactly `propext`, `Classical.choice`, `Quot.sound` for all 7, and the kernel types `theorem_1_1` as `breakdownStatement`. The record decides one field, whether the energy clause is a formal conclusion, at both commits by one rule, and refuses unless the rule calls `8937a8f4` not formal, as the audit read it; it does, and it calls `f9e8bc5b` formal. The 11 files the reading rests on are pinned by digest with corpus copies, and the probe battery re-decides the field from those copies with 2 red controls of its own.

**Is it the paper's theorem?** Comparator does not judge this statement; it judges (C) and (D), which are unchanged. So `breakdownStatement` was read the way Section 3.1 reads those, clause by clause (Table 3; 11 rows, 9 *same* and 2 *equivalent*). Theorem 1.1 is an existence statement with a negation attached, so the direction rule applies twice: in the existence part a clause weaker than the paper's (a junk value, a missing integrability) would make the theorem cheap, and in the negation a competitor class stronger than the paper's would. Neither occurs. Unlike (C) and (D), whose statement a third party wrote months before the proof, this statement is OpenAI's own: it is in `R3/ProblemStatement.lean`, byte-identical at the pin, where it stood unproved under its own docstring. Its authority is this reading, not Comparator's.

**What it does not change.** Comparator judges (C) and (D), not Theorem 1.1, and `formalization.yaml` still declares the same four main results: the energy clause is formal in a theorem the repository proves and through which its proof of (C) passes, not in a statement a third party wrote. Comparator was not re-run at `f9e8bc5b`, its binaries being among those that abort on this machine; its acceptance of both configurations on 2026-09-09 is a fact about `8937a8f4`. At `f9e8bc5b` the challenge files and the solution-side definitions are byte-identical to the pin, so the judged statements are the same, and the new proofs were checked by Lean's kernel as they compiled and their axioms asked; nanoda did not replay them. For the two days between the announcement and `f9e8bc5b` the energy clause was a paper claim, and the register keeps that as the row's history. Unchanged by the commit and still standing: the growth is the pointwise form with no rate, $h$ is a number in neither artifact, and every asserted step of Section 3.4 is about the paper.

## 5 What the record decides

The register row `navier-stokes-openai-2026`, recorded 2026-10-05, reads CERTIFIED: at `f9e8bc5b` the Lean proves the paper's Theorem 1.1 as stated, finite energy included, and derives Clay's (C) and (D) from it. It keeps as history the PARTIAL recorded 2026-09-09 at `8937a8f4`, the announced commit: the Lean certificate held for Clay's (C) and (D), and the finite-energy clause the paper's Theorem 1.1 states was not in the formal statement [19]. Exactly:

1. *Held.* The formal statements are Clay's (C) and (D) with nothing added to the competitor class and nothing dropped from the data, written by a third party before the proof existed; the competitor classes are inhabited (Section 3.1).

2. *Held, at `8937a8f4`.* All 2,486 modules build on this machine with the pinned toolchain; the kernel reports exactly `propext`, `Classical.choice`, `Quot.sound` for each of the 4 main declarations; Comparator accepts both configurations through Lean's kernel and nanoda (Section 3.2).

3. *Held, at `8937a8f4`.* Every clause of Theorem 1.1 except the uniform energy bound and the vanishing of the force near $t = 0$ has a formal counterpart, and the formal uniqueness covers every competitor Clay admits (Section 3.3). The two exceptions were paper claims at that commit, the first resting on the paper's Lemma 10.4, re-derived by hand.

4. *Held, at `f9e8bc5b`.* Both exceptions are conclusions of `theorem_1_1`, whose statement reads as the paper's Theorem 1.1 clause by clause, 9 *same* and 2 *equivalent*; the 486

changed and dependent modules build here over the pinned build, and the kernel reports exactly `propext`, `Classical.choice`, `Quot.sound` for all 7 declarations asked (Section 4).

5. *Held, as readings.* No error, circularity or counterexample was found in six independent readings; 6 of the 12 asserted steps are discharged in Lean at `8937a8f4` at full strength, 2 more strongly than the paper states; no classical obstruction refutes the construction; all 7 probe scripts pass with all 26 red controls firing; the Euler transfer step reads complete (Sections 3.4–3.8).

6. *Not decided.* At `f9e8bc5b` Comparator was not re-run and nanoda did not replay the new proofs; the statement of Theorem 1.1 that `theorem_1_1` proves is OpenAI's own, so its fidelity is this note's reading, not a third party's statement (Section 4).

7. *Not decided.* The smallness parameter $h$ is bounded numerically in neither artifact; the proof is asymptotic in a parameter no one has quantified, and nothing here makes it effective.

8. *Not decided.* In the trace at `8937a8f4`, not repeated at `f9e8bc5b`, the instantiation of Proposition B.2's machinery for the actual profile was not traced to a line and Lemma 4.9's collar inequality has no formal counterpart; the paper names no sentence for the swirl obstruction it evades; the 3 Euler checks are listed and unrun; Comparator's process isolation did not run.

9. *Not decided.* The mathematics of the writeup beyond what the kernel and the readings establish. An audit is not a referee report, and the readings were not peer review.

## 6 What is not claimed

Nothing is claimed about the correctness of the mathematics beyond items 1–5 above: the kernel establishes that the proof terms check against the third party's statement, the readings that nobody here found an error, and the verdict on the theorem is the build's and the kernels', not the reading's. No numerical computation here certifies anything: the paper contains no numerics and none of its constants is a number. Nothing is claimed about priority: the record holds the dates each party states (Section 7) and adjudicates nothing, and the concurrent results [10] are different theorems from the one audited here. Nothing is claimed about the prize: the papers are published and not peer-reviewed, the Clay Mathematics Institute had said nothing at the time of the record, and OpenAI has said it does not intend to claim it [4]. The word *certified* is used here only for what a kernel checked.

## 7 The dates, as the parties state them

The record holds 12 dates, each from the document named in it: OpenAI's announcement [4], Buckmaster's statement and the concurrent papers [10], Tao's posts [11], or a repository's own timestamps [5, 6]. Nothing here weighs them [19].

**2026-05-15**: DeepMind's Formal Conjectures adds the Lean statement of the four Clay alternatives (#1457); **2026-07-27**: last change to that file (#4643, divergence lemmas); the revision OpenAI pinned; **2026-08-15**: Alpöge–Buckmaster obtain smooth-forcing blowup for Boussinesq and Euler (their statement); **2026-08-22**: Alpöge–Buckmaster's Lean verification (their statement); **2026-08-28**: OpenAI begins training the internal model used (the announcement); **2026-09-01**: OpenAI launches the agent effort after 'a rumor that two Millennium

Prize problems had been resolved' (the announcement); **2026-09-03**: Tao's six-part post on the opportunity cost of an autonomous solution; Buckmaster writes to OpenAI (his statement); **2026-09-05**: OpenAI's agents 'arrive at their resolution', 88 hours in (the announcement); Tao's clarification: 'not aware of any significant developments'; **2026-09-06**: Lean formalization complete, 17 hours later (the announcement); the two calls with Buckmaster (his statement); **2026-09-08 03:58 UTC**: Alpöge–Buckmaster publish IPM, Boussinesq and Euler with smooth forcing, Lean at tristanbuckmaster/fluid_lean (repository created 04:03 UTC); **2026-09-08 10:53 UTC**: openai/NavierStokesAndEuler created; its single commit at 10:57 UTC; the announcement the same day; **2026-09-10 11:51 UTC**: openai/NavierStokesAndEuler's commit f9e8bc5b (message ".", Boris Alexeev): Theorem 1.1 proved with its energy bound, (C) and (D) re-derived from it (the repository's own timestamps; read here 2026-10-05).

## Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The records are in `corpus/navier-stokes/` [19]: `lean-repo.json` is computed from a clone at `8937a8f4` by `tools/pin-navierstokes-lean.js`, `build.json` by `tools/record-navierstokes-build.js`, and `probes.json` by `python3 instruments/navierstokes/battery.py`, which re-runs the 7 probe scripts with their 26 red controls. For the upstream commit, `python3 tools/build-navierstokes-head.py` rebuilds `f9e8bc5b` over the pinned build and `node tools/record-navierstokes-upstream.js` writes `upstream-f9e8bc5b.json` from a clone at that commit and the build's log, decides the energy field at both commits and asks the kernel; its reading is the `upstream` key of `audit.json`. The macros of this document are written by `node tools/paper-numbers/navier-stokes-audit.js` from those records at repository commit `9349d8bad505`; the timing fields of `probes.json`, which change on every run, are never read. Rebuilding the Lean repository needs the clone and toolchain the tools name and about 140 minutes on an eight-core machine, the upstream commit 25 more; nothing here depends on re-running either, and every number above is read from the records of the runs of 2026-09-09 and 2026-10-06.

## References


[1] C. L. Fefferman, Existence and smoothness of the Navier–Stokes equation, Clay Mathematics Institute, 2000, with errata, `https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf`, pinned 2026-09-09.

[2] OpenAI, Finite time blowup for Navier–Stokes, 166 pp., 8 September 2026, `https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf`, pinned sha256 `0e779481`…

[3] OpenAI, Finite time blowup for the Euler equation, 57 pp., 8 September 2026, `https://cdn.openai.com/pdf/315b36cd-ec98-4023-8342-93345194ece1/euler.pdf`.

[4] OpenAI, On the Navier–Stokes Millennium Prize Problem, 8 September 2026, `https://openai.com/index/navier-stokes-solution/`, pinned 2026-09-09 via the Internet Archive.

[5] OpenAI, NavierStokesAndEuler: Lean 4 formalizations, `https://github.com/openai/NavierStokesAndEuler`, commit `8937a8f4`, 2026-09-08 (the announced commit), and commit `f9e8bc5b`, 2026-09-10.

[6] Google DeepMind, Formal Conjectures, `FormalConjectures/Millenium/NavierStokes.lean`, `https://github.com/google-deepmind/formal-conjectures`, revision `8bf45ed7`; the statement added 2026-05-15 in #1457.

[7] L. de Moura, S. Ullrich, The Lean 4 theorem prover and programming language, in: Automated Deduction – CADE 28, LNCS 12699, Springer, 2021, 625–635; The mathlib Community, The Lean mathematical library, in: Proc. CPP 2020, ACM, 367–381; Mathlib revision `85e3a25e` as pinned.

[8] Lean FRO, Comparator, `https://github.com/leanprover/comparator`, revision `19e111e2`; lean4export, `https://github.com/leanprover/lean4export`, revision `cacf989b`; nanoda (`ammkrn/nanoda_lib`), a Lean kernel written in Rust, built from source for the run.

[9] D. Córdoba, L. Martínez-Zoroa, Finite time singularities of smooth solutions for the 2D incompressible porous media (IPM) equation with a smooth source, arXiv:2410.22920v3 (2025).

[10] L. Alpöge, T. Buckmaster, Blowup for the Boussinesq equations with smooth forcing, and Blowup for the Euler equations with smooth forcing; with M. P. Coiculescu, Extending the Córdoba–Martínez-Zoroa IPM blow-up to uniformly space-time smooth forcing; T. Buckmaster, statement of 8 September 2026; all at `https://cims.nyu.edu/~tristanb/`, pinned 2026-09-09.

[11] T. Tao, posts of 3 and 5 September 2026, `https://mathstodon.xyz/@tao/117207849921390904`, pinned 2026-09-09.

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[19] cert-machine, the records of the audit in `corpus/navier-stokes/` (`MANIFEST.json`, `lean-repo.json`, `build.json`, `nonvacuity.json`, `fidelity-clay-vs-lean.md`, `dissection.md`, `audit.json`, `euler.md`, `probes.json`), of 2026-09-09; the reading of the upstream commit, `upstream-f9e8bc5b.json` (2026-10-06) with the 11 files it pins in `lean/f9e8bc5b/` and the `upstream` key of `audit.json` (2026-10-05); the register row `navier-stokes-openai-2026` in `certs/claims-ledger.json`; and the page built from them, `https://carlostoledo.co/reports/navier-stokes.html`.

| item | the paper | the Lean | status |
|---|---|---|---|
| the two theorems | Theorem 1.1 (C), Corollary 10.6 (D) | `navier_stokes_breakdown_R3`, `_periodic`, through `ComparatorBridge` adapters; solution-side definitions byte-identical to the challenge | proved, subject to the build record |
| uniqueness, Lemma 10.5 | $v$ smooth, $L^\infty_t L^2_x$, zero datum, same force $\Rightarrow v = u$; no growth condition on the pressure | `classical_uniqueness_on_Icc` (`R3/WholeSpaceUniqueness.lean:30`): competitor smooth, divergence-free, uniformly finite energy, pressure only smooth; pressure recovered (`R3/PressureRecovery.lean:419`), flux bounded uniformly in the radius (`R3/PressureFlux.lean:576`) | covers every competitor Clay admits |
| the viscosity rescaling | (10.22): space rescaled, singular time 1 | time rescaled, $f_\nu(x,t) = \nu^2 f(\nu t, x)$ (`ComparatorR3Theorem.lean:19`), singular time $1/\nu$ | equivalent; the statement names no time |
| the uniform energy bound | Theorem 1.1's $\sup_{t<1}\|u(t)\|_{L^2} < \infty$; Lemma 10.4; the announcement's "its energy remains finite through the entire dynamics" | `uniform_finite_energy` (`R3/CompactEnergy.lean:343`) proves it for any smooth compactly supported force and is used by nothing; the structure carrying `energy_bounded` has no producer; the candidate's `Properties` structure has ten fields and no energy field | NOT FORMAL at `8937a8f4`; every other clause of Theorem 1.1 is; formal at `f9e8bc5b` |
| the force's support in time | $f \in C^\infty_c(\mathbb{R}^3 \times (0,\infty))$: zero near $t = 0$ | smooth for $t \geq 0$, compact spatial support, zero for $t \geq T$; no field states $f = 0$ near $t = 0$ | weaker in Lean at `8937a8f4`; the challenge's force condition does not require it; formal at `f9e8bc5b` |
| the growth | (10.20)–(10.21): $u_\theta = \tau^{-A}(e_0 + O(\tau^{2h}))$ along a named off-axis path | `SpeedUnboundedAtOne`: $\forall M\ \delta\ \exists t\, x$, $1 - \delta < t < 1$, $M < \|u(t,x)\|$; on the torus also $\|u(t,0)\| \to \infty$ on the axis | pointwise $\limsup\|u\|_\infty = \infty$; no rate |
| the numbering | Propositions 5.5, 7.5, 9.6, 9.9, Theorem 3.1 | docstrings cite a manuscript numbering absent from the published paper | the map is by content |

Table 2: The paper mapped onto the certificate at the announced commit `8937a8f4`, from the record [19]; the two rows not formal there are formal at the upstream commit `f9e8bc5b` (Section 4).

| Theorem 1.1 | `breakdownStatement` at `f9e8bc5b` (`R3/ProblemStatement.lean`) | direction |
|---|---|---|
| for every $\nu > 0$ | `∀ ν : ℝ, 0 < ν → …` | same |
| a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty);\mathbb{R}^3)$ | `force_smooth : ContDiff ℝ ∞ f` on all of $\mathbb{R}\times\mathbb{R}^3$; `force_support : CompactPositiveTimeSupport f`: compact support, `tsupport f ⊆ Ioi 0 ×ˢ univ` | same: the zero extension of a $C_c^\infty$ field on the open half-space |
| a compact $K \subset \mathbb{R}^3$ | `support_compact : IsCompact K` | same |
| $u, p$ smooth on $\mathbb{R}^3 \times [0,1)$ | `ContDiffOn ℝ ∞ u preSingularDomain`, `preSingularDomain = Ico 0 1 ×ˢ univ`; likewise $p$ | same, as Clay's (6) |
| $\partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p = f$ | `navier_stokes`: for $0 < t < 1$ and every $x$ the residual equals $f(t,x)$; `fderiv` in $t$, $\sum_i \partial_i\partial_i$, $\nu$ on the Laplacian only | equivalent: on the open slab every derivative is genuine; $t = 0$ follows by continuity |
| $\nabla\cdot u = 0$ | `divergence_free`, for $0 \le t < 1$ | same |
| $u(\cdot,0) = 0$ | `zero_initial_velocity` | same |
| $\operatorname{supp} u(\cdot,t) \cup \operatorname{supp} p(\cdot,t) \subset K$, $0 \le t < 1$ | `velocity_support`, `pressure_support`: the closed support (`tsupport`) in $K$ | same |
| $\sup_{0\le t<1}\|u(t)\|_{L^2} < \infty$ | `energy_bounded : UniformFiniteEnergy (Ico 0 1) u`: $\exists E \ge 0$, $\forall t \in [0,1)$, $\|u(t,\cdot)\|^2$ integrable and $\frac{1}{2}\int\|u(t,\cdot)\|^2 \le E$ | same: integrability is stated, so the integral's junk value 0 cannot meet the bound vacuously |
| $\limsup_{t\uparrow 1}\|u(t)\|_{L^\infty} = \infty$ | `speed_unbounded : SpeedUnboundedAtOne u`: $\forall M > 0$, $\forall\delta > 0$, $\exists t \in (0,1)$, $1-\delta < t$, $\exists x$, $M < \|u(t,x)\|$ | equivalent for continuous slices; no rate |
| no smooth global $(u,P)$ with the same force and datum and $\sup_{t\ge 0}\frac{1}{2}\int\|u\|^2 < \infty$ | `¬ Nonempty (GlobalFiniteEnergySolution ν f)`: smooth for $t \ge 0$, zero datum, divergence-free, the equation for $t > 0$ with the same $\nu$ and $f$, `UniformFiniteEnergy (Ici 0)` | same: no pressure growth, support or decay is asked, so the negation is not weakened |

Table 3: The paper's Theorem 1.1 against `breakdownStatement`, the statement `theorem_1_1` proves at `f9e8bc5b`, one row per clause (11 rows; 9 read *same*, 2 *equivalent*). Transcribed from the record [19].